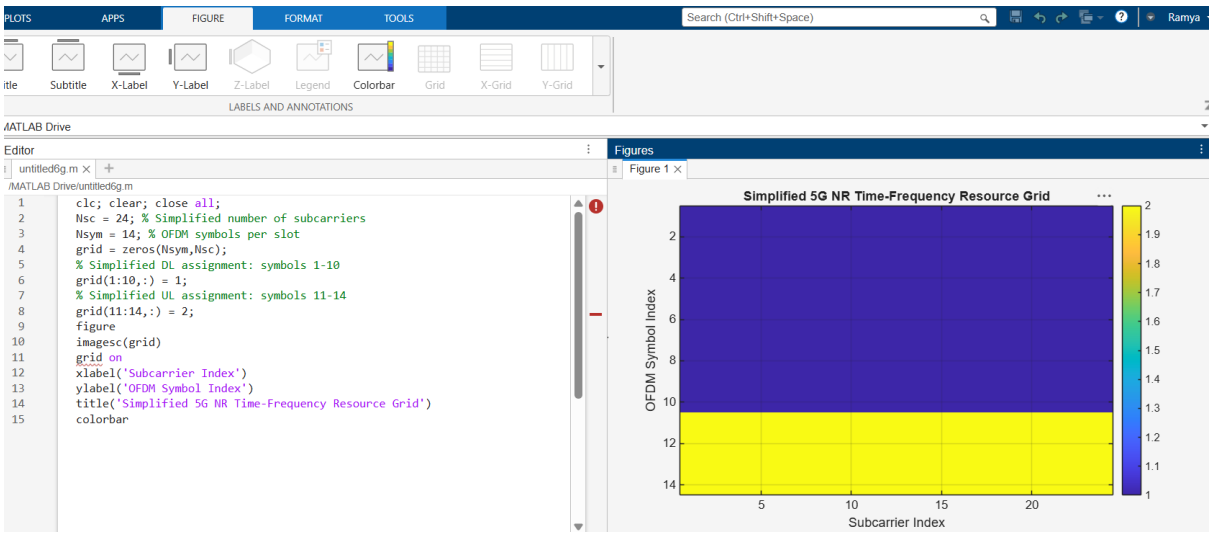
**Experiment: 7**

**Analysis of 5G NR Frame Structure and Subcarrier Spacing**

**Objective – 1: Generate / Inspect a Simplified 5G NR Resource Grid**

**Program:**

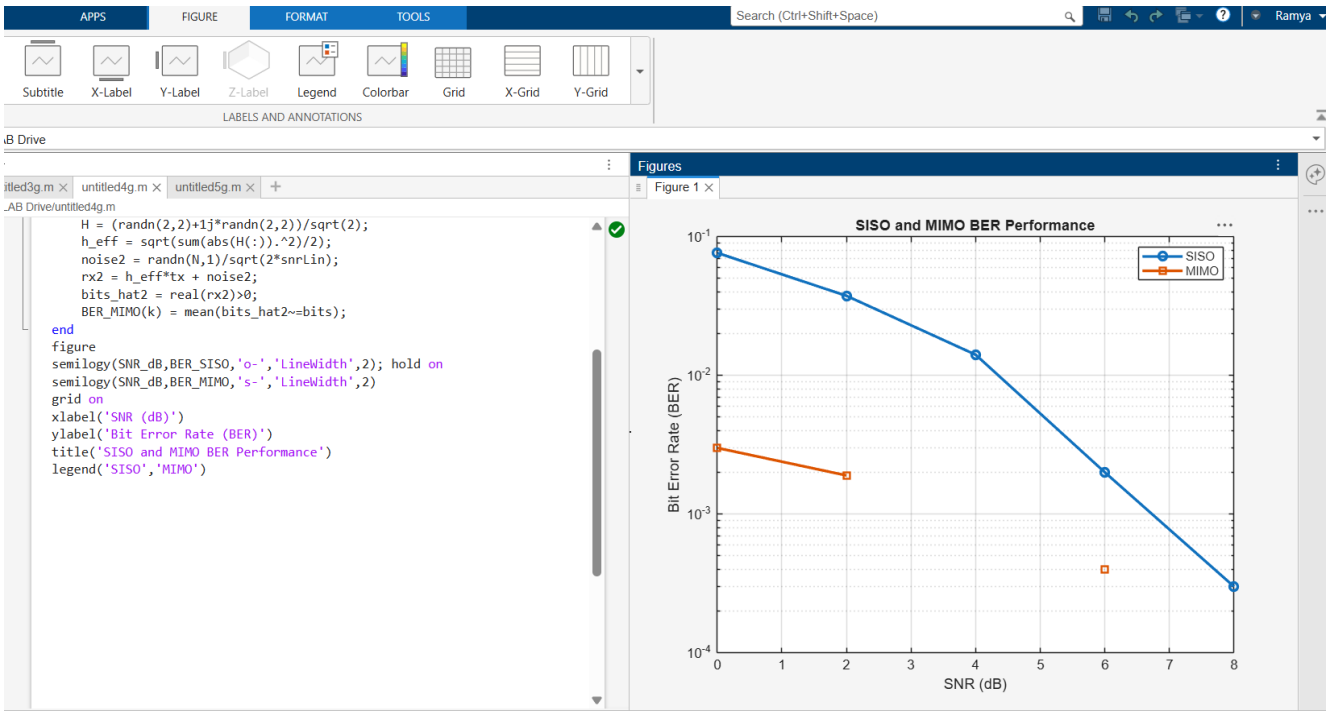
```
clc; clear; close all;
Nsc = 24; % Simplified number of subcarriers
Nsym = 14; % OFDM symbols per slot
grid = zeros(Nsym,Nsc);
% Simplified DL assignment: symbols 1-10
grid(1:10,:) = 1;
% Simplified UL assignment: symbols 11-14
grid(11:14,:) = 2;
figure
imagesc(grid)
grid on
xlabel('Subcarrier Index')
ylabel('OFDM Symbol Index')
title('Simplified 5G NR Time-Frequency Resource Grid')
colorbar
```



**Objective – 2: Compare SCS and Slot Duration**

**Program:**

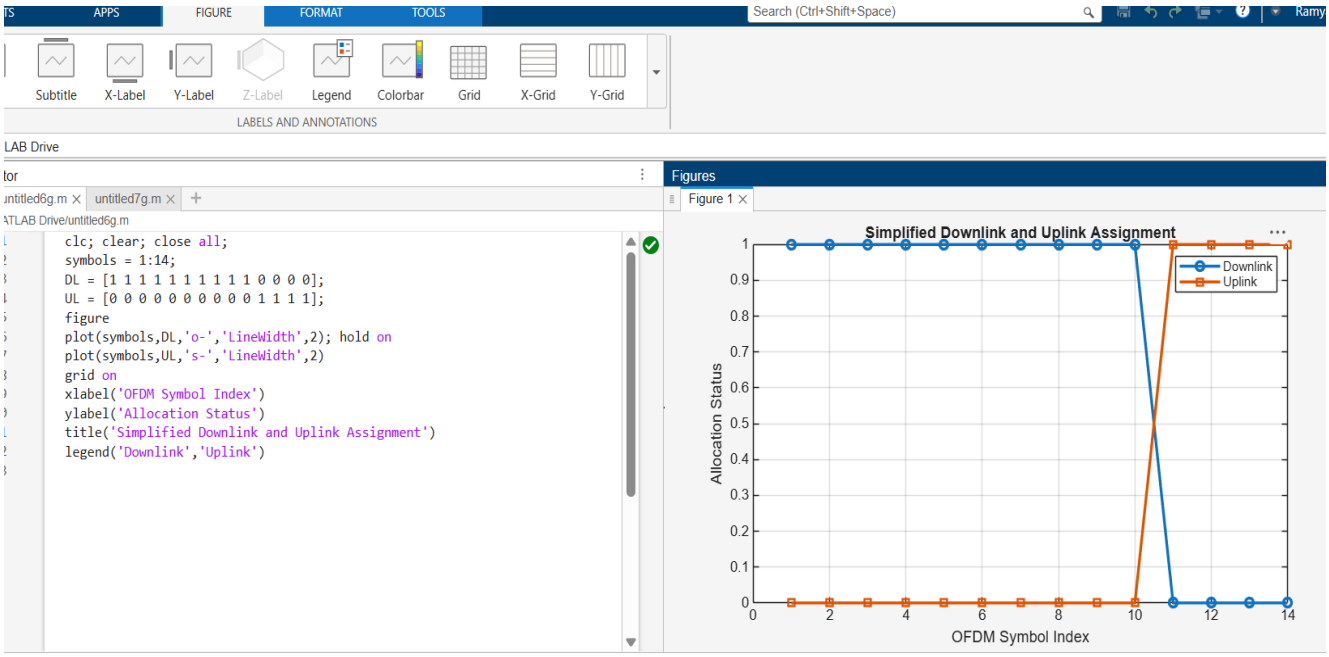
```
clc; clear; close all;
scs = [15 30 60 120 240]; % kHz
mu = 0:4;
slot = [1 0.5 0.25 0.125 0.0625]; % ms
figure
plot(scs,slot,'o-','LineWidth',2)
grid on
xlabel('Subcarrier Spacing (kHz)')
ylabel('Slot Duration (ms)')
title('5G NR Slot Duration for Different Subcarrier Spacing')
figure
bar(scs)
grid on
set(gca,'XTickLabel',mu)
xlabel('Numerology Index (\mu)')
ylabel('Subcarrier Spacing (kHz)')
title('5G NR Subcarrier Spacing for Different Numerologies')
```




Objective – 3: Identify Basic DL and UL Resource Assignments

**Program:**

```
clc; clear; close all;
symbols = 1:14;
DL = [1 1 1 1 1 1 1 1 1 1 0 0 0 0];
UL = [0 0 0 0 0 0 0 0 0 0 1 1 1 1];
figure
plot(symbols,DL,'o-','LineWidth',2); hold on
plot(symbols,UL,'s-','LineWidth',2)
grid on
xlabel('OFDM Symbol Index')
ylabel('Allocation Status')
title('Simplified Downlink and Uplink Assignment')
legend('Downlink','Uplink'
```


APPS
FIGURE
FORMAT
TOOLS
Search (Ctrl+Shift+Space)
Subtitle
X-Label
Y-Label
Z-Label
Legend
Colorbar
Grid
X-Grid
Y-Grid
LABELS AND ANNOTATIONS
untitled6g.m
untitled7g.m
ATLAB Drive/untitled6g.m
clc; clear; close all;
symbols = 1:14;
DL = [1 1 1 1 1 1 1 1 1 1 0 0 0 0];
UL = [0 0 0 0 0 0 0 0 0 0 1 1 1 1];
figure
plot(symbols,DL,'o-','LineWidth',2); hold on
plot(symbols,UL,'s-','LineWidth',2)
grid on
xlabel('OFDM Symbol Index')
ylabel('Allocation Status')
title('Simplified Downlink and Uplink Assignment')
legend('Downlink','Uplink')
Figures
Figure 1
Simplified Downlink and Uplink Assignment
Downlink
Uplink
Allocation Status
OFDM Symbol Index
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
2
4
6
8
10
12
14